Question ID: 2b89bfe5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Medium

Question

While researching a topic, a student has taken the following notes:

- In 1999, astronomer Todd Henry studied the differences in surface temperature between the Sun and nearby stars.
- His team mapped all stars within 10 parsecs (approximately 200 trillion miles) of the Sun.
- The surface temperature of the Sun is around 9,800°F, which classifies it as a G star.
- 327 of the 357 stars in the study were classified as K or M stars, with surface temperatures under 8,900°F (cooler than the Sun).
- 11 of the 357 stars in the study were classified as A or F stars, with surface temperatures greater than 10,300°F (hotter than the Sun).

The student wants to emphasize how hot the Sun is relative to nearby stars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. At around 9,800°F, which classifies it as a G star, the Sun is hotter than most but not all of the stars within 10 parsecs of it.

B. Astronomer Todd Henry determined that the Sun, at around 9,800°F, is a G star, and several other stars within a 10-parsec range are A or F stars.

C. Of the 357 stars within ten parsecs of the Sun, 327 are classified as K or M stars, with surface temperatures under 8,900°F.

D. While most of the stars within 10 parsecs of the Sun are classified as K, M, A, or F stars, the Sun is classified as a G star due to its surface temperature of 9,800°F.

Correct Answer: A

Rationale

Choice A is the best answer. Noting that the Sun (9,800°F) is hotter than most stars within 10 parsecs of it, the sentence emphasizes how hot the Sun is relative to nearby stars.

Choice B is incorrect. The sentence explains that astronomer Todd Henry determined the classifications for the Sun and several other stars nearby; it doesn’t emphasize how hot the Sun is relative to nearby stars. Choice C is incorrect. The sentence explains that the majority of stars near the Sun are classified as K or M stars; it doesn’t indicate the Sun’s temperature or emphasize how hot it is relative to nearby stars. Choice D is incorrect. While the sentence indicates that the Sun is classified differently than most nearby stars due to its surface temperature, it doesn’t emphasize how hot the Sun is relative to nearby stars.